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Nutritional and anti-nutritional evaluation of injera made from Sorghum, Rice, Teff, and Flaxseed flours using D-optimal mixture design

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Injera, a fermented staple food traditionally made from *Teff* flour, is a key part of the Ethiopian diet, with nearly all Ethiopians eating it at least once daily. Compositing various grain-based, and oilseed flours with *Teff* flour improves the nutritional content, and cost-effectiveness of the produced *Injera*. In Ethiopia, very limited studies were conducted to investigate the use of sorghum, rice, and *teff* flour blends combined with flaxseed flour in *Injera* preparation. Thus, this study was conducted to evaluate the anti-nutritional factors, microbiological characteristics, and ideal blending proportions of sorghum, rice, *teff*, and flaxseed flours to create a superior quality *Injera* via D-optimal design. One-way and two-way ANOVA, alongside Tukey's pairwise comparison method, was employed to assess mean differences. The aim was to enhance the nutritional quality of *Injera* compared to a control sample made from 100% sorghum flour. The optimal nutritional profile was achieved with a blend comprising 50% sorghum, 20% rice, 30% *teff*, and supplemented with 3% flaxseed flour. This blend resulted in *Injera* containing 3.02% fiber, 9.30% protein, 83.16% carbohydrate, 393.24 kcal/100 g energy, 13.99 mg/100 g iron, 4.06 mg/100 g calcium, and 3.78 mg/100 g zinc. The optimization process identified this blend as optimal for improving the nutritional quality of *Injera*. The findings underscore that blending *teff* with sorghum, rice, and flaxseed flour enhances the nutritive value of *Injera*.

Keywords D-optimal, *Injera*, Nutritional profile, Blending

Injera is a traditional flatbread that serves as a dietary staple in Ethiopia and surrounding regions¹. It is known for its unique spongy texture and slightly tangy taste². This flatbread is typically made from *teff* flour, which comes from a highly nutritious grain³. *Teff* is valued for its high protein content, dietary fiber, and essential minerals like iron and calcium⁴. The grain *Teff* (*Eragrostis tef* (Zucc.) Trotter) has a nutritional composition of approximately 11% protein, 73% carbohydrates, 3% crude fiber, 2.5% fat, and 2.8% ash³. *Teff* flour is also utilized in food products, including breads, cookies, and baby foods. This is due to its whole grain usage, gluten-free properties, high iron content, and the digestibility of its proteins and starches^{3,5}.

Currently, *Teff* is the priciest cereal cultivated in Ethiopia, largely because *Injera* made from *Teff* is a popular and culturally significant dish, often seen as a status symbol within the society (predominantly consumed in urban areas where household incomes are higher)^{3,6}. Additionally, *Teff* and *Injera* are exported to countries like North America, the Middle East, and Europe due to their gluten-free nature and nutritional benefits^{6,7}.

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Despite these advantages, there is growing interest in boosting the nutritional value of *Injera* while lowering its production costs by blending *Teff* with other flours, such as sorghum, rice, and flaxseed^{2,8,9}. These alternatives are not only rich in essential nutrients but are also more widely available³.

Sorghum (*Sorghum bicolor* (L.) Moench.) is a vital cereal in some semiarid regions of the world, with around 40% of global production used for human consumption¹⁰. It was reported to have a proximate composition of 74.7% carbohydrates, 1.8% ash, 12.3% protein, 4.2% fat, and 1.7% fiber^{3,11}. Sorghum is used as the second most preferred cereal for *Injera* preparation in Ethiopia. However, there has been minimal research on improving sorghum *Injera* quality¹². Meanwhile, using Rice (*Oryza sativa* L.) flour in *Injera* production has gained popularity as a cost-effective alternative. Some modern recipes incorporate a mixture of *teff* and rice flour^{13,14}. A common blend used in these recipes is around 50% *teff* flour and 50% rice flour^{15,16}. Other ratio variations might be used, such as 70% *teff* flour to 30% rice flour, depending on availability and desired nutritional balance^{17,18}. *Injera* made from a blend of cereals provides enhanced nutrition, offering higher levels of protein, iron, and calcium (notably with *teff*) while retaining good fiber content. *Injera* made entirely from sorghum is gluten-free and rich in antioxidants but has less accessible iron, lower calcium content, and incomplete protein. Nutritionally, rice is rich in beneficial phytochemicals, fiber, vitamins, minerals, and antioxidants, with lower amylose and higher amylopectin content contributing to better freeze-thaw stability^{19,20}. Flaxseed (*Linum Usitatissimum*), known for its health benefits and nutritional value, enhances dough uniformity, and crumb elasticity, and prolongs the freshness of baked goods by improving water-binding capacity and reducing moisture loss^{21,22}. Incorporating flaxseed into *Injera* enhances the omega-3 fatty acids, lignans, and dietary fiber, which can offer additional health benefits, including improved heart health and digestive support^{1,23,24}. Compositing these grains flours with *teff* flour in various proportions has the potential to create a more nutritionally diverse product, improving the protein, fat, and fiber content of *Injera*¹⁸.

Composite flours enhance the nutrient profile, and overall appeal of *Injera*, and could significantly contribute to food security²⁵. Yet, the use of composite flours raises concerns about anti-nutritional components like phytic acid, tannins, and phenols²⁶. These substances can bind to minerals and reduce their absorption, thereby decreasing the bioavailability of essential nutrients^{27,28}. Therefore, it is important to assess these anti-nutritional factors to ensure that the improved nutritional qualities of the composite *Injera* can be effectively absorbed and utilized by the body.

However, there is limited research on using blends of sorghum, rice, and *teff* flours without flaxseed flour in *Injera* preparation²⁹. Given that sorghum and rice are more affordable in Ethiopia, there is a growing interest in mixing these ingredients with *teff* flour^{1,29}. This research aims to encounter the rising demand for *Teff* together with the enhancement of the nutritional profile of *Injera* produced, by thoroughly investigating the nutritional, anti-nutritional, and microbiological aspects with optimal blending ratios of sorghum, rice, *teff*, and flaxseed flours to produce higher-quality *Injera*.

Materials and methods

Sample collection

The experimental materials included 15 kg of white sorghum (*Melkam* variety) obtained from the Gonder Agricultural Research Center, 14 kg of brown rice (*X-Jigena* variety) from the Fogera National Rice Research and Training Center, 8 kg of white *Teff* (*Qoncho Teff* variety, DZ-Cr-387) from Adet Agricultural Research Center, and 2 kg of flaxseed (*Geregera* variety). These specific varieties were chosen due to their widespread popularity, extensive production, and frequent use by Ethiopian farmers.

Sample preparation

The samples collected were prepared by established methods of^{1,22,30–32}. Each ingredient was processed according to the procedure outlined in Fig. 1. The sorghum, rice, *teff*, and flaxseed grains were cleaned manually through winnowing and handpicking. The sorghum, rice, and *teff* grains were milled into whole flour using a community cottage disk miller and then sieved using 710 µm and 600 µm sieves, respectively, based on their effectiveness^{13,30,32}. Flaxseed was also ground using a high-speed multi-function oilseed and coffee mill. The prepared samples were stored in polyethylene plastic bags at a refrigeration temperature of 4 °C until further experimental analysis.

Treatment combinations and experimental design

The experiment utilized a D-optimal mixture design, incorporating three independent variable ingredients. The lower and upper treatment combination ranges were 43 to 50% for Sorghum flour, 20 to 27% Rice flour, and 23 to 30% *Teff* flour. Blending ratios for sorghum, rice, and *teff* flours were determined through preliminary trials and previous research^{11,18,22}, and then calculated using Minitab software version 19.2. Each experimental treatment included a constant 3% flaxseed flour, except for treatments with 100% sorghum or *teff* flour.

Composite flour preparation

The composite flour was prepared according to the generated experimental design for *injera* preparation. The *injera* was prepared with 100% sorghum flour and 100% *teff* flour considered as the control^{12,33}, and the composition is shown in Table 1.

Batter preparation, fermentation, and *Injera* preparation

The *Injera* was made using a specific composite flour blend of 0.6 kg (600 g) mixed with 1 L (1000 mL) of water to form a thick dough. Fermentation was initiated by adding 30 mL of *Ersho*, a starter culture saved from a preliminary test batter^{1,12}. The dough was kneaded for 5 min to ensure uniformity, then sealed and fermented at room temperature for 72 h. Afterward, 100 mL of the surface liquid was removed, and 200 mL of potable

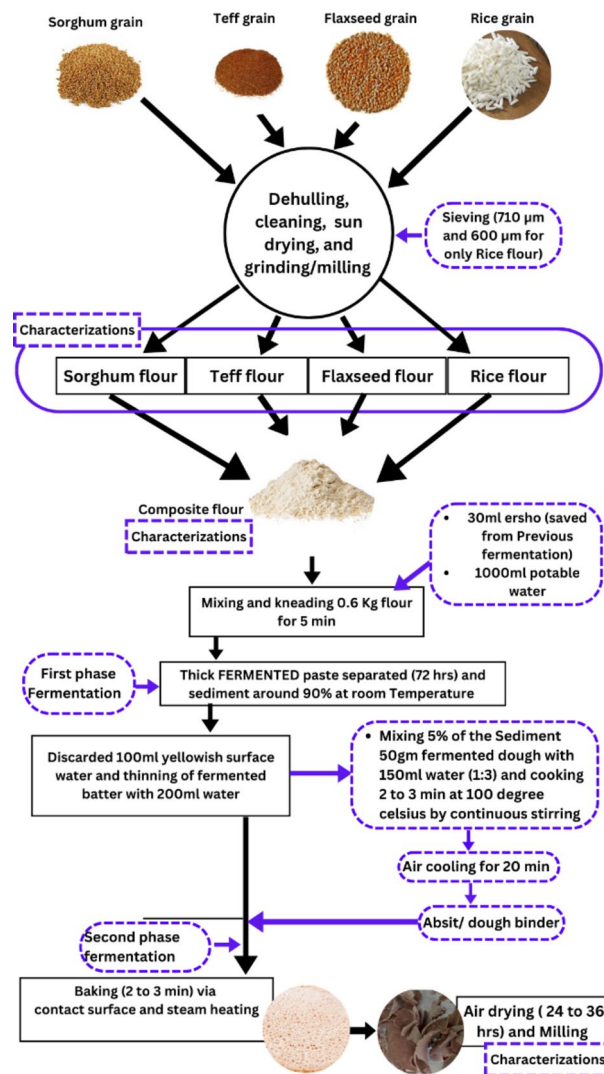


Fig. 1. The experimental framework of the study design.

Sample code	Blocks	Sorghum flour (%)	Rice flour (%)	Teff flour (%)	Supplement flaxseed flour (%)
Con1	1	100	0	0	0
Con2	1	0	0	100	0
BR1	1	50	27	23	3
BR2	1	50	20	30	3
BR3	1	46.5	23.5	30	3
BR4	1	46.5	27	26.5	3
BR5	1	43	27	30	3
BR6	1	47.67	24.67	27.66	3
BR7	1	45.34	25.83	28.83	3
BR8	1	48.83	22.34	28.83	3
BR9	1	48.83	25.83	25.34	3
BR10	1	50	23.5	26.5	3

Table 1. Formulations of blending ratios with constant supplemented flaxseed flour. *Con1 = 100% sorghum flour, con2 = 100% *teff* flour, BR = blending ratio.

water was added to thin the dough. To make *Absit*, 5% of the fermented dough was mixed with water (1:2 ratio), cooked at 100 °C for 3 min to gelatinize the starch, cooled, and reintroduced into the fermenting dough.

The second phase of fermentation lasted for 16 h, marked by bubble formation indicating that the fermentation was completed and the dough was ready for baking. Finally, 100 ml of potable water was added to the fermented dough to achieve the desired batter consistency^{30,32}. About 500 g of fermented batter were poured in a circular motion onto a hot griddle called *Metad* (45–50 cm in diameter) and covered with a bamboo-based lid, *Akambalo*, to trap steam. Before pouring, the surface was rubbed with rapeseed flour using cloth. The *Injera* was baked at 190–210 °C for 2–3 min, then removed and stored in an airtight *Mesob*, a traditional woven container^{1,12,30,32,34,35}. For determination (analysis) of proximate, mineral, and anti-nutritional properties the proportion of *Injera* was air-dried for 24–36 h at room temperature with the controlled conditions and converted into powder by FW100 high-speed universal disintegrator. After the analysis, the results were reported in dry base^{32,36}.

Proximate composition of *Injera*

Moisture content

The moisture content of freshly baked *Injera* from composite flour was analyzed using the oven drying method (AOAC 2000, method 930.15). Pre-dried and weighed moisture dishes (W1) were filled with 5 g samples (W2) in triplicate and dried at 105 °C for 6 h until reaching a constant weight. After cooling in a desiccator, the final weight (W3) was recorded. Then, the moisture content was estimated by Eq. (1).

$$\text{Moisture content (MC\%)} = \frac{W2 - W3}{W2 - W1} \times 100. \quad (1)$$

Crude protein

Protein content was measured using the Kjeldahl method, specifically the 920.87 protocol for analyzing crude protein in flour³⁷. Approximately 0.5 g of the food sample was weighed and placed in a digestion flask. The sample was digested at 50 to 415 °C for 190 min with 5 mL of concentrated H₂SO₄ (as an oxidizing agent), anhydrous Na₂SO₄ (to raise the boiling point and speed up the reaction), and 1 gram of a catalyst mixture (Na₂SO₄ and CuSO₄ in a 10:1 ratio). This process converted nitrogen (excluding nitrates and nitrites) into ammonium sulfate, while organic matter was broken down into CO₂ and H₂O. After digestion, the mixture was diluted with 50 mL of distilled water and 40 mL of 40% NaOH was added to release ammonia gas. The sample was distilled into a flask containing 25 mL of 4% boric acid until the volume reached 150 mL. The resulting borate ion was titrated with standard 0.1 N HCl to complete the analysis.

$$\text{Total nitrogen} = \frac{(T - B) \times N \times 14.007 \times 100}{W} \quad (2)$$

where T-Volume in mL of the standard acid solution used in the titration for the test material, B-Volume in mL of the standard acid solution used in the titration for the blank determination, N-Normality of standard sulfuric acid, and W-Weight in grams of the test material.

$$\text{Crude protein} = \text{conversion factor} \times \text{total nitrogen} \quad (3)$$

Conversion factor (5.7) for sorghum, rice, and *teff* cereals and (5.3) for flaxseed³⁸.

Crude fat

Crude fat was analyzed using the Soxhlet extraction method (AOAC, 2000; method 920.39). A 5 g sample was placed in filter paper and extracted with hexane in a Soxhlet apparatus for 6–8 h. The extract was transferred to a pre-weighed evaporating beaker, rinsed with hexane, and evaporated until odorless. The beaker was dried at 120 °C for 30 min, cooled in a desiccator, and re-weighed to determine the fat content. The Crude fat content was determined by Eq. (4).

$$\text{Crude fat (\%)} = \frac{W2 - W1}{\text{Weight of sample}} \times 100, \quad (4)$$

where: W₁ = Weight of extraction flask before extraction, and W₂ = Weight of extraction flask after extraction.

Crude fiber

The crude fiber content was determined using a non-enzymatic gravimetric method as described by AOAC (2000). A 2 g sample of defatted food was placed in a 150 mL beaker with 60 mL of 1.25% H₂SO₄ and a pre-weighed 2 g boiled sample. The mixture was heated on a digestion apparatus, boiled for 45 min with periodic shaking, and then filtered through a screen sieve. The beaker was rinsed three times with 50 mL of near-boiling water, and the rinses were filtered. The residue was transferred to a digestion flask with 40 mL of 1.25% NaOH and boiled for 30 min, shaking every 5 min. The final residue was filtered using a 75 µm crucible under a 25 mm vacuum, dried at 130 °C for 2 h, cooled in desiccators, and weighed (M₁). The dried residue was ignited for 2 h at 600 ± 15 °C until ashing was completed and then cooled in desiccators and reweighed (M₂). The crude fiber was determined with Eq. (5).

$$\text{Crude fiber (\%)} = \frac{M_1 - M_2}{\text{Weight of sample}} \times 100, \quad (5)$$

where M_1 = mass of crucible and residue before ignition, and M_2 = mass of crucible and residue after ignition.

Total ash

The ash content was performed by the official method 942.05 of AOAC (2000). The crucibles were thoroughly cleaned, dried at 120°C , and pre-ignited at 550°C for 3 h. After cooling in desiccators, their mass was measured (M_1) using an analytical balance. Approximately 5 g of the sample was weighed into the crucibles (M_2) and heated in a furnace at 550°C for about 8 h until carbon was removed and the residue turned grayish-white. The sample was removed from the furnace placed in the desiccators and weighed (M_3), the ash content was analyzed by Eq. 6.

$$\text{Total ash (\%)} = \frac{M_3 - M_1}{M_2 - M_1} \times 100. \quad (6)$$

Total carbohydrate

The total amount of carbohydrates in the samples was calculated by subtracting the parameters measured in the aforementioned test from 100%³⁷.

$$\text{CHO (\%)} = 100\% - (\%M + \%CP + \%A + \%CF) \quad (7)$$

where; % M (DB) is the percent of moisture content in dry base, %CP is the percent of crude protein, %A is the percent of ash, and %CF is the percent of crude fat.

Energy value/calorific value

The calorific value of food (in kcal/100 g) was determined by multiplying each gram of protein, fat, and carbohydrate obtained from laboratory analysis by their respective conversion factor.

$$\text{Energy} \left(\frac{\text{kcal}}{100\text{g}} \right) = (\text{protein} \times 4) + (\text{carbohydrate} \times 4) + (\text{fat} \times 9). \quad (8)$$

The energy content of the food sample is determined by using multiplication factors of 4, 9, and 4 for every gram of protein, fat, and carbohydrate, respectively^{39,40}.

Determination of anti-nutritional factors

Phytate

The phytate content was determined by the method outlined by⁴¹, with modifications by⁴². Approximately 0.1 g of dried Injera flour were extracted with 10 mL of 0.2% HCl at room temperature for one hour. After centrifugation at 3000 rpm for 30 min, the clear supernatant was used for phytate analysis. For the assay, 3 mL of the sample was mixed with 2 mL of Wade reagent (0.06% $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ and 0.6% sulfosalicylic acid), and absorbance was measured at 500 nm using a UV-VIS spectrophotometer. Phytate concentration was calculated using a standard curve (4–32 ppm) and expressed as phytic acid in mg per 100 g of dry weight.

$$\text{Phytic acid} \left(\frac{\text{mg}}{100\text{g}} \right) = \frac{(\text{AB} - \text{AS}) - \text{intercept} \times 10}{\text{slope} \times \text{weight of sample} \times 3} \quad (9)$$

Where AB is the absorbance of the blank sample, AS is the absorbance of the sample, and 3 is the volume extracted from the sample (supernatant).

Condensed tannin

The method outlined by⁴³ was employed for determining condensed tannins. A 0.30 g sample was placed in a screw-cap test tube with 8 mL of 1% HCl in methanol. The mixture was vortexed for 10 s and incubated in a water bath for 20 min, with a 10-second centrifugation halfway through. After incubation, the sample was vortexed again for 10 s, centrifuged at 1000 rpm for 15 min, and the supernatant was carefully decanted to avoid transferring solid residues. Two 1 mL aliquots were taken from the supernatant and placed into separate test tubes. One tube served as the sample, and the other as the blank. Both tubes were placed in a 30°C water bath for 20 min. During this period, 5 mL of vanillin reagent was added to the sample tube and 5 mL of 4% HCl in methanol was added to the blank tube at 15-second intervals. After 20 min, the absorbance of each sample and blank tube was measured at 500 nm, with the spectrophotometer zeroed using a methanol blank. 40 mg of D-Catechin was weighed and dissolved in 100 mL of 1% HCl in methanol (stock) solution, then taken 0, 2.69, 5.38, 26.88, 53.76, and 107.52 ppm of stock solution was in a test tube. Adjust the volume of each tube to 1 mL with 1% HCl in methanol.

$$\text{Tannin in} \left(\frac{\text{mg}}{100\text{g}} \right) = \frac{V \times A/m}{W} \quad (10)$$

Where V is the volume of extract in mL. A is Absorbance at 500 nm (absorbance of sample-absorbance of blank). m is the slope of the standard curve. W is the Weight of the sample.

Total phenolic content

Phenolic compounds concentration in the *injera* flour was extracted by methanolic extracts as described by⁴⁴. For each sample, 5 g were homogenized with 10 mL of extraction solvent comprising 60% methanol and 0.1% HCl (acidic aqueous methanol by volume: H₂O: HCl, 70:29.5:0.5) and heated in a water bath at 85 °C for 2 h. After cooling, the mixture was transferred to a 100 mL volumetric flask and diluted to volume with distilled water. The extract was filtered under slight pressure using Whatman filter paper. A 5 mL aliquot of the extract was mixed with 15 mL distilled water, 5 mL Folin-Ciocalteu reagent, and 10 mL saturated sodium carbonate solution, then diluted to 100 mL with distilled water. The reaction was allowed to proceed for 30 min at room temperature, and absorbance was measured at 750 nm using a UV-Vis spectrophotometer. A standard curve was prepared using gallic acid solutions at 0, 0.05, 0.1, 0.15, and 0.2 ppm. The total content of phenolic compounds in sample extracts and gallic acid equivalent (GAE) was calculated by the formula:

$$TP = \left(\frac{C \times V}{m} \right) \quad (11)$$

where TP is the total phenolic content in mg/g, in GAE (Gallic acid equivalent), C is the concentration of gallic acid established from the calibration curve in mg/mL, V is the volume of sample extract in mL, and m is the weight of the sample in g.

Total flavonoid content

Total flavonoid content was determined by a colorimetric method as described by⁴⁵. Total flavonoid content was measured by mixing 1 g of *injera* flour extract or 100 µL of supernatant with methanol and 2% AlCl₃ solution. Absorbance was recorded at 415 nm after 10 min, using a blank without AlCl₃ for calibration. Flavonoid levels were quantified using a quercetin standard curve (0–0.1 ppm) and expressed as mg quercetin equivalents (QE) per g of dry extract, based on the mean of three readings.

$$\text{Total flavonoid} \left(\frac{\text{mg}}{100\text{g}} \right) = \frac{C \times V}{W} \quad (12)$$

Where C is the concentration of quercetin equivalents (QE) per g of extract established from the calibration curve in mg/mL, V is the volume of solvent extract in mL, and W is the weight of the sample in g.

Mineral analysis

The mineral content of calcium, iron, zinc, magnesium, manganese, and potassium was analyzed using atomic absorption spectrophotometry (AAS) and flame photometry, following AOAC (2000) standards. A 2 g milled sample was digested in a 2:5 mixture of concentrated nitric acid (69%) and hydrochloric acid (37%) and then diluted to 50 mL with distilled water. Standard stock solutions for each mineral (iron, zinc, calcium, magnesium, manganese, and potassium) were prepared from 10 ppm stock solutions. The samples and standards were atomized using an air-acetylene flame, and absorbance was measured at specific wavelengths: 248.33 nm for iron, 213.8 nm for zinc, 422.67 nm for calcium, 285.21 nm for magnesium, 279.48 nm for manganese, and 766.49 nm for potassium. Then the solution was used to determine Fe, Zn, Ca, Mg, Mn, and K elements in the flour and formulated *injera* samples by Eq. 13.

$$\text{Mineral content} \left(\frac{\text{mg}}{100\text{g}} \right) = \left(\frac{A - B}{W} \right) \times V \times D \quad (13)$$

Where W is the Weight (g) of the sample, V is the Volume of extract in mL, A is the Concentration (mg/L) of the sample solution, B is the concentration (mg/L) of the blank solution and D is the Dilution factor.

Total microbial load analysis

For microbiological stability analysis, freshly baked *Injera* samples were cooled for 1–2 h, then wrapped in clean polyethylene plastic and stored in a traditional *Messob* basket at room temperature (around 25 °C) for 6 days. Samples were randomly collected on days 0, 2, 4, and 6 using sterile materials. The total aerobic bacteria and yeast-mold counts were determined using AACC (2000) methods, and results were reported as colony-forming units (CFU) per gram. Yeast mold and total bacteria were determined using the pour plate technique as described by²². To determine yeast and mold count, potato dextrose agar (PDA) was poured into plates, incubated aerobically at 25 °C for 3–5 days, and colonies were counted (30–300 colonies per plate) as colony-forming units per gram of *Injera* (CFU/g). For total bacterial count, plate count agar (PCA) was used, and after incubating the plates aerobically at 35 °C for 48 h, total aerobic bacteria were enumerated on plates with 30–300 colonies using a colony counter (CFU/g).

Statistical data analysis and optimal validation

All experimental measurements were conducted in triplicate, except for microbial load, which was done in duplicate. Data analysis was performed using Minitab version 19.2 with statistical methods including linear

and quadratic regression models for mixtures, single-factor analysis, and two-way ANOVA. Tukey’s pairwise comparison was used to assess the significance of mean values at a 5% significance level ($p < 0.05$).

Results and discussions
Proximate composition of formulated *Injera*

Moisture content

The moisture content of the blended baked *Injera* ranged from 60.03 to 64.31%. In contrast, the moisture content of the control *Injera*, made solely from sorghum and *teff* flour, was 59.46% and 63.05%, respectively. This suggests that the blending ratio affects the moisture variation and softness of *Injera* under controlled baking conditions. As reported by⁴⁶, *injera* made from various cereals has a moisture content exceeding 50%. Additionally³⁵, and⁴⁷ indicated that the moisture content of blended *injera* products can reach 62–65%.

The lowest moisture content (60.03%) was found in the *injera* with BR2 (50% sorghum, 20% rice, and 30% *teff*), while the highest moisture content (64.31%) was observed in the *injera* with BR3 (46.5% sorghum, 23.5% rice, and 30% *teff*). As detailed in Table 2, the moisture content of the formulated *injera* decreased with higher sorghum flour composition and significantly increased with more *teff* and rice. Despite the initial moisture content of sorghum and *teff* flour being not significantly different ($p < 0.05$), sorghum *injera* experienced greater moisture loss during storage compared to *teff injera*¹². This study consistently substituted 3% of flaxseed flour for each blended ratio⁴⁸.

According to^{23,49}, flaxseed flour supplementation in cereal products positively affects moisture retention during storage after baking due to its water-binding potential. The interaction between the ingredients in the formulated *injera* showed a significant difference ($p < 0.05$) and high significance ($p < 0.05$) between the quadratic model interactions of sorghum with rice, sorghum with *teff*, and rice with *teff* ($R^2 = 0.9476$) (Table 2). The predicted moisture content of the formulated *injera* is estimated by Eq. 14.

$$mc = 1.5625*sorghum - 10.5955*rice + 4.7381*teff + 0.1606*sorghum*rice + 0.1349*sorghum*teff + 0.1572*rice*teff \quad (14)$$

Total ash content

In this study, ash content values were recorded in the range of 1.82–2.40% in the blend *injera*. Whereas sorghum and *teff* flour *injera* ash values are 1.70% and 2.31% respectively. The formulated blended *injera* BR4, which consists of 46.5% sorghum, 27% rice, and 26.5% *teff*, along with a 3% flaxseed flour supplement, showed the lowest total ash content at 1.82%. In contrast, the highest total ash content of 2.40% was found in the blended *injera* sample BR5, made up of 43% sorghum, 27% rice, 30% *teff*, and a 3% flaxseed flour substitution. As the composition of rice and sorghum flour increased the ash content also decreased but the *teff* flour composition increased the ash composition linearly increased (Eq. 15). The analyzed result showed that the ash content of *injera* optimization was significantly affected ($p < 0.05$) through the blending of different cereal components (Table 3). According to^{11,30,50} reports *injera* prepared from blended flours with higher *teff* flour compositions shows a higher ash content due to its high amount of minerals and ash in *teff* flour from other cereals. The predicted linear and quadratic models between sorghum, rice, and *teff* were significantly different among sorghum control *injera* ($R^2 = 0.9587$).

$$\text{Ash content} = -0.12069*sorghum - 1.02769*rice + 1.97975*teff + 0.03141*sorghum*rice - 0.03153*sorghum*teff - 0.02500*rice*teff \quad (15)$$

Crude fat content

The crude fat levels of *injera* formulated with a 3% flaxseed flour supplement ranged from 2.50 to 3.20%. In contrast, the crude fat content of 100% sorghum and *teff* flour *injera* (used as controls) were 2.02% and 1.98%, respectively. There was a highly significant difference ($p < 0.05$) in the crude fat percentages observed in the interaction of sorghum with rice, sorghum with *teff*, and rice with *teff*, as shown by the linear and quadratic models ($R^2 = 0.9733$) (Table 2).

The lowest fat content (2.5%) was found in BR4 *injera*, made from a blend of 46.5% sorghum, 27% rice, and 26.5% *teff*. Conversely, the highest crude fat content (3.20%) was observed in BR2 *injera*, consisting of 50% sorghum, 20% rice, and 30% *teff*. The presence of sorghum in these blends increased the crude fat content of the

Regression	MC	TA	CF	CF*	CP	CHO	Energy
Linear model	0.008	0.002	0.009	0.000	0.005	0.016	0.000
Quadratic model	0.014	0.003	0.002	0.000	0.002	0.034	0.000
Sorghum*rice	0.017	0.004	0.004	0.000	0.043	0.031	0.000
Sorghum*teff	0.030	0.004	0.006	0.001	0.003	0.030	0.130
Rice*teff	0.018	0.010	0.002	0.039	0.001	0.151	0.000
R ²	0.9476	0.9587	0.9733	0.9938	0.9707	0.9704	0.9942

Table 2. ANOVA mixture compositions P-value proximate analysis. *Mc = moisture content, TA = total ash, CF = crude fat, CF* = crude fiber, CP = crude protein, CHO = carbohydrate, R² = determination of correlation coefficient.

Compositions (S: R: T)	Moisture content (%)	Total ash (%)	Crude fat (%)	Crude fiber (%)	Crude protein (%)	Total carbohydrate (%)	Energy/calorific value (kcal/100 g)
CON1	59.46 ± 0.51 ^d	1.70 ± 0.106 ^d	2.02 ± 0.03 ^{cd}	3.69 ± 0.01 ^b	8.91 ± 0.14 ^d	83.68 ± 0.13 ^b	388.33 ± 0.32 ^d
CON2	63.05 ± 0.32 ^{ab}	2.31 ± 0.25 ^{abc}	1.98 ± 0.02 ^d	2.83 ± 0.22 ^{de}	8.14 ± 0.01 ^e	84.74 ± 0.24 ^a	389.38 ± 1.01 ^d
BR1	60.11 ± 0.13 ^{cd}	2.01 ± 0.23 ^{abcd}	3.07 ± 0.16 ^{ab}	3.00 ± 0.17 ^{cd}	9.20 ± 0.11 ^{bc}	83.50 ± 0.55 ^b	393.30 ± 1.66 ^{bc}
BR2	60.03 ± 0.05 ^{cd}	1.92 ± 0.10 ^{abcd}	3.20 ± 0.05 ^a	2.40 ± 0.31 ^e	9.31 ± 0.04 ^a	83.67 ± 0.64 ^b	398.26 ± 0.79 ^a
BR3	64.31 ± 0.63 ^a	2.34 ± 0.33 ^{ab}	3.17 ± 0.05 ^a	2.40 ± 0.22 ^e	9.18 ± 0.07 ^{abc}	82.91 ± 0.15 ^{bc}	396.89 ± 0.82 ^{ab}
BR4	60.07 ± 0.06 ^{cd}	1.82 ± 0.08 ^d	2.50 ± 0.32 ^{bcd}	4.50 ± 0.16 ^a	9.01 ± 0.01 ^{bcd}	81.61 ± 0.15 ^{de}	390.05 ± 1.69 ^{cd}
BR5	63.00 ± 0.04 ^{ab}	2.40 ± 0.14 ^a	2.89 ± 0.07 ^{ab}	4.50 ± 0.27 ^a	9.20 ± 0.10 ^{ab}	81.39 ± 0.42 ^e	388.40 ± 1.20 ^d
BR6	62.22 ± 0.75 ^b	2.04 ± 0.15 ^{abcd}	2.60 ± 0.51 ^{abc}	3.00 ± 0.13 ^{cd}	9.00 ± 0.02 ^{cd}	82.75 ± 0.28 ^{bc}	395.80 ± 0.72 ^{ab}
BR7	62.35 ± 0.16 ^b	1.99 ± 0.09 ^{abcd}	3.06 ± 0.01 ^{ab}	3.50 ± 0.22 ^{bc}	9.03 ± 0.08 ^{bcd}	82.43 ± 0.34 ^{cd}	393.35 ± 1.18 ^{bc}
BR8	62.10 ± 0.19 ^b	2.00 ± 0.10 ^{abcd}	3.03 ± 0.28 ^{ab}	2.50 ± 0.25 ^{de}	9.05 ± 0.00 ^{bcd}	83.43 ± 0.46 ^{bc}	397.18 ± 1.66 ^a
BR9	61.55 ± 1.36 ^{bc}	1.89 ± 0.10 ^{bcd}	3.01 ± 0.16 ^{ab}	3.00 ± 0.09 ^{cd}	9.00 ± 0.03 ^{cd}	83.09 ± 0.20 ^{bc}	395.48 ± 0.90 ^{ab}
BR10	62.73 ± 0.74 ^{ab}	1.86 ± 0.03 ^{cd}	3.06 ± 0.10 ^{ab}	2.40 ± 0.15 ^e	9.01 ± 0.04 ^{bcd}	83.17 ± 0.18 ^{bc}	393.29 ± 1.97 ^{bc}
P-value	0.011	0.007	0.003	0.000	0.004	0.004	0.000

Table 3. Proximate analysis of sorghum: rice: *teff* formulated *injera*. *Con1–100% sorghum flour, con2–100% *teff* flour, BR - blending ratio, S – sorghum flour, R – rice flour, T – *teff* flour. *Each value means ± standard deviation of triplicates. The values followed by different superscript letters within a column indicate significant differences ($p < 0.05$).

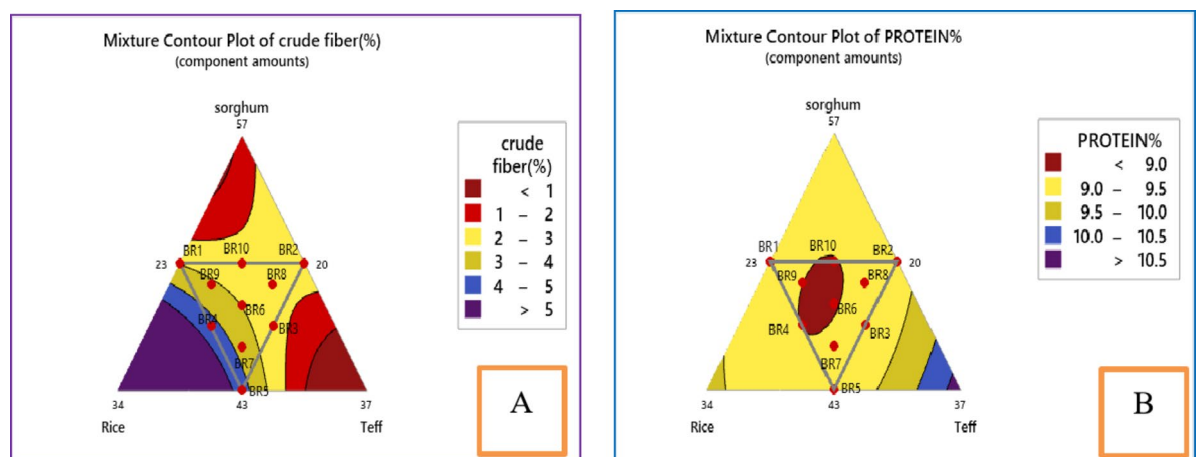


Fig. 2. Contour plot of crude fiber (A) & crude protein interaction (B) response.

injera. This increase is attributed to the relatively high fat content found in sorghum flour (2.10–7.60%,⁵¹), and rice flour (1.6–2.8 mg/100g^{52,53}), .

$$\begin{aligned} \text{Crude fat} = & -0.50724 * \text{sorghum} - 1.55159 * \text{rice} - 1.23141 * \text{teff} + 0.03114 * \text{sorghum} * \text{rice} \\ & + 0.02736 * \text{sorghum} * \text{teff} + 0.03960 * \text{rice} * \text{teff} \end{aligned} \quad (16)$$

Crude fiber content

The crude fiber content of formulated *injera* ranged from 2.40 to 4.50%, as shown in Table 3. The highest crude fiber content (4.50%) was observed in *injera* made from blending ratios BR4 (46.5% sorghum, 27% rice, 26.5% *teff*) and BR5 (43% sorghum, 27% rice, 30% *teff*). Conversely, the lowest crude fiber content (2.40%) was found in *injera* from blending ratios BR2 (50% sorghum, 20% rice, 30% *teff*), BR3 (46.5% sorghum, 23.5% rice, 30% *teff*), and BR10 (50% sorghum, 23.5% rice, 26.5% *teff*). *Injera* made with 100% sorghum flour (control) had a crude fiber content of 3.69%, which was higher than that of *injera* made with 100% *teff* flour (control), which had a content of 2.83%. There was a significant difference ($p < 0.05$) in crude fiber content between sorghum and *teff* flours. The crude fiber content in the blended *injera* samples was significantly influenced by the addition of brown rice flour, followed by sorghum flour, as the amount of flaxseed flour remained constant in each blend (Fig. 2). Although *teff* grain is known for its high bran and fiber content³⁰, brown rice and sorghum flours also have high fiber content, as reported by⁵⁴. The crude fiber content of composite flour *injera* samples varied from 0.8 to 1.0% and 3.91–5.2%, respectively^{18,55}.

$$\begin{aligned} \text{Crude fiber model} = & 0.00748 \cdot \text{sorghum} + 6.08400 \cdot \text{rice} - 3.70515 \cdot \text{teff} - 0.11436 \cdot \text{sorghum} \cdot \text{rice} \\ & + 0.08156 \cdot \text{sorghum} \cdot \text{teff} - 0.02644 \cdot \text{rice} \cdot \text{teff} \end{aligned} \quad (17)$$

The crude fiber content of *injera*, based on the formulated blending ratio, exhibited a highly significant difference in both the linear and quadratic models when sorghum was combined with rice flour and *teff* flour. The model showed a significant difference ($p < 0.05$). The coefficient of determination for the formulated model was exceptionally high ($R^2 = 0.9938$).

Crude protein content

The crude protein content of the formulated *injera* ranged from 9.00 to 9.31%, while the sorghum and *teff* control *injera* products contained 8.91% and 8.14% crude protein, respectively (Table 3). The protein content of all formulated *injera* products showed a highly significant difference ($p < 0.05$) in both linear and quadratic models when sorghum was combined with *teff* flour and rice with *teff* flour. Significant variation was also observed in the interaction between sorghum and rice. The highest relative protein content (9.31%) was found in *injera* prepared with the BR2 blend (50% sorghum, 20% rice, 30% *teff*), while the lowest relative protein content (9.00%) was found in the BR6 (47.67% sorghum, 24.67% rice, 27.66% *teff*) and BR9 (48.83% sorghum, 25.83% rice, 25.34% *teff*) blends.

The increase in protein content of the formulated *injera* was attributed to the higher percentages of sorghum and *teff*, despite the constant supplementation of flaxseed, which has a high protein content. However, the contour plot and model predictions indicated that the protein content of the formulated *injera* was more influenced by *teff* flour, followed by rice flour (Fig. 2 and Eq. 18). The observed increase in protein content may be due to the improved or optimized protein content from blending and composite flours^{18,22} and could be linked to enzyme synthesis or changes in compositional proportions followed by the degradation of components⁵⁶. Despite the constant addition of flaxseed, its protein content remained highly significant. Previous studies reported mean composite flour *injera* protein content in the range of 7.47–14.74%, which aligns with findings of^{13,18,50,57}. The coefficient of determination for the interaction model was $R^2 = 0.9707$.

$$\begin{aligned} \text{Protein content} = & 0.276329 \cdot \text{sorghum} + 0.577828 \cdot \text{rice} + 0.973288 \cdot \text{teff} - 0.007667 \cdot \text{sorghum} \cdot \text{rice} \\ & - 0.016570 \cdot \text{sorghum} \cdot \text{teff} - 0.021244 \cdot \text{rice} \cdot \text{teff} \end{aligned} \quad (18)$$

Total carbohydrate content

The carbohydrate content of formulated *injera* ranged from 81.39 to 83.67%, with sorghum and *teff* flour *injera* control values being 83.68% and 84.74%, respectively. The highest carbohydrate content (83.67%) was found in *injera* BR2 (50% sorghum, 20% rice, 30% *teff*), while the lowest (81.39%) was in BR5 (43% sorghum, 27% rice, 30% *teff*) (Table 3). There was a significant difference ($p < 0.05$) in the linear and quadratic models of carbohydrate interaction, such as between sorghum and rice and sorghum and *teff* flour (Table 2). The carbohydrate content of the blended baked *injera* increased as the proportion of *teff* flour, followed by sorghum flour, and rose (Fig. 3 & Eq. 19). Studies by^{11,33} indicate that sorghum, rice, and *teff* flour are rich sources of carbohydrates. *Injera* made from different composite flours showed a mean total carbohydrate content ranging from 73.89 to 86.8%, aligning with the findings of^{57,58}.

$$\begin{aligned} \text{Total carbohydrate} = & 1.32599 \cdot \text{sorghum} + 3.0618 \cdot \text{rice} + 2.97857 \cdot \text{teff} + 0.05962 \cdot \text{sorghum} \cdot \text{rice} \\ & - 0.06016 \cdot \text{sorghum} \cdot \text{teff} + 0.03224 \cdot \text{rice} \cdot \text{teff} \end{aligned} \quad (19)$$

The interaction significance model coefficient determination value ($R^2 = 0.9704\%$).

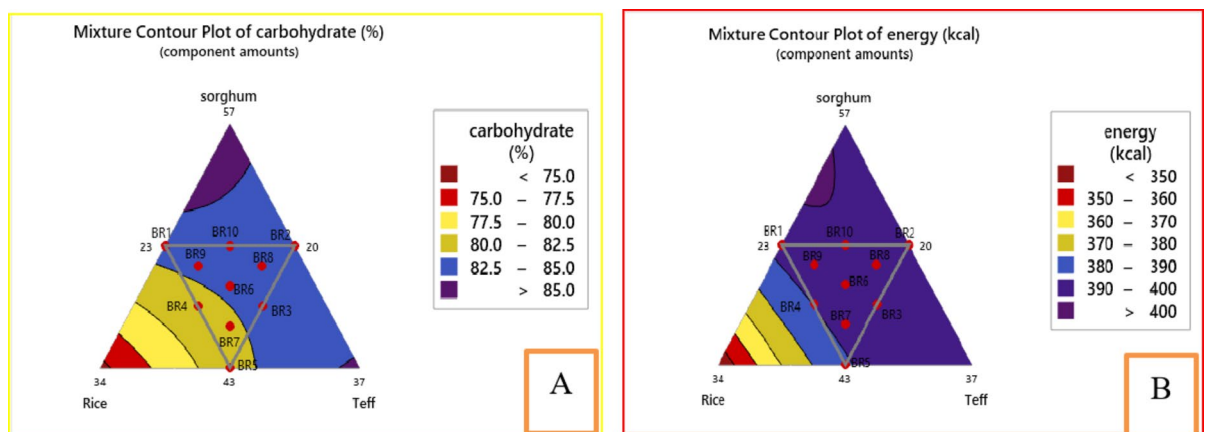


Fig. 3. Mixture contour plot of carbohydrate (A) & gross energy interaction (B) response.

Gross energy value

The gross energy value of the formulated *injera* ranged from 388.40 to 398.26 kcal/100 g, with the control samples of 100% sorghum and *teff* flour *injera* showing energy values of 388.40 and 398.26 kcal/100 g, respectively (Table 3). The highest energy content was found in *injera* made from the BR2 blend (50% sorghum, 20% rice, 30% *teff*), while the lowest was in the BR5 blend (43% sorghum, 27% rice, 30% *teff*). There was a highly significant difference ($p < 0.05$) in the caloric content of the composite flour *injera* in both linear and quadratic models involving sorghum with rice, and rice with *teff*, whereas the difference between sorghum and *teff* was not significant (Table 2). Flaxseed flour did not significantly affect the gross energy of *injera* due to its low inclusion rate. These findings align with^{11,59}, who reported that the gross energy of composite flour *injera* varied from 349.60 to 380.25 kcal/100 g. The increased energy value in the formulated *injera* is attributed to the high carbohydrate content in the sorghum, rice, and *teff* flour used in the study (Fig. 3).

Energy model equation = 1.8525*sorghum - 24.0421*rice + 4.7608*teff + 0.4903*sorghum*rice - 0.0621*sorghum*teff + 0.4026*rice*teff (20)

Anti-nutritional factor analysis of composite flour injera

The phytate content in *injera* made from various composite flours ranged from 175.66 to 177.93 mg/100 g. Specifically, the phytate levels in 100% sorghum and *teff* flours were 178.26 mg/100 g and 176.14 mg/100 g, respectively. The highest phytate content (177.93 mg/100 g) was found in the BR7 blend (45.34% sorghum, 25.83% rice, 28.83% *teff*), while the lowest phytate content (175.66 mg/100 g) was observed in the BR9 blend (48.83% sorghum, 25.83% rice, 25.34% *teff*). Although *teff* flour had a lower phytate content compared to sorghum flour, an increase in the proportion of *teff* in the composite flour resulted in higher phytate concentrations in the *injera*. Table 4 illustrates the levels of anti-nutritional factors, total phenols, and total flavonoid content in *injera* samples.

The phytate content in *injera* made from blends aligns with the findings reported by^{11,51,60}. The phytate content in *injera* made from *teff* flour specifically also corresponds with the results reported by¹¹. The study showed no significant difference ($p < 0.05$) in the interaction between linear and quadratic models regarding the mixture of components, likely because all components in the mixture are rich sources of phytate. This anti-nutritional content chelates and binds minerals, reducing their bioavailability for nutritional absorption^{1,61}, but also provides health benefits, such as anti-inflammatory and anti-cancer properties^{24,30}. According to the model (Eq. 21) and the mixture contour plot (Fig. 4), the phytate content increases as the proportion of *teff* flour, followed by rice and sorghum, increases.

Phytate content = 1.11458*sorghum + 1.62260*rice + 2.91276*teff + 0.02397*sorghum*rice + 0.00023*sorghum*teff - 0.03712*rice*teff (21)

The amount of Total Phenol (TP) content ranged from 3.03 to 5.39 mg/100 g, and the phenol amount of individual sorghum and *teff* flours *injera* was 4.07 mg/100 g and 4.70 mg/100 g, respectively. The highest phenol content (5.39 mg/100 g) was determined in the blended *injera* prepared from BR2 (50% sorghum, 20% rice, 30% *teff*), whereas the lowest phenol content (3.03 mg/100 g) was present in BR5 (43% sorghum, 27% rice, 30% *teff*). The blended *injera* product total phenol result indicated that similar to the mixture contour plot and predicted model response composition of sorghum flour was increased the total phenol concentration in the blended *injera* also increased. According to⁶² studies the total phenol concentration of sorghum flour *injera* resulted in 3.69 mg/100 g and it was in agreement with the current study result. Different works of literature

Compositions	Total phenol (mg/100 g)	Condensed tannin (mg/100 g)	Phytate (mg/100 g)	Total flavonoid (mg/100 g)
CON1	4.07 ± 0.39 ^{bcd}	0.04 ± 0.08 ^a	178.26 ± 1.09 ^a	289.90 ± 0.65 ^c
CON2	4.70 ± 0.18 ^{abc}	0.00 ± 0.00 ^a	176.14 ± 0.35 ^{ef}	304.90 ± 4.19 ^{abc}
BR1	3.68 ± 0.33 ^{cd}	0.04 ± 0.15 ^a	176.34 ± 0.07 ^{def}	368.00 ± 1.31 ^{abc}
BR2	3.03 ± 0.83 ^d	0.00 ± 0.13 ^a	177.70 ± 0.78 ^{abc}	371.60 ± 5.85 ^{abc}
BR3	3.72 ± 0.37 ^{cd}	0.04 ± 0.20 ^a	177.49 ± 0.00 ^{abcd}	361.50 ± 6.30 ^{abc}
BR4	4.10 ± 0.55 ^{abcd}	0.13 ± 0.13 ^a	176.59 ± 0.34 ^{cdef}	349.70 ± 1.80 ^{abc}
BR5	5.39 ± 0.26 ^a	0.09 ± 0.08 ^a	176.92 ± 0.09 ^{bcddef}	298.50 ± 5.89 ^{bc}
BR6	5.10 ± 0.09 ^{ab}	0.04 ± 0.15 ^a	177.05 ± 0.07 ^{abcde}	388.20 ± 7.40 ^a
BR7	5.30 ± 0.28 ^{ab}	0.09 ± 0.20 ^a	177.93 ± 0.07 ^{ab}	364.00 ± 5.40 ^{abc}
BR8	4.35 ± 0.48 ^{abc}	0.17 ± 0.20 ^a	176.93 ± 0.15 ^{bcddef}	364.80 ± 2.75 ^{abc}
BR9	5.16 ± 0.31 ^{ab}	0.04 ± 0.08 ^a	175.66 ± 0.02 ^f	351.30 ± 3.80 ^{abc}
BR10	5.22 ± 0.69 ^{ab}	0.00 ± 0.26 ^a	176.70 ± 0.42 ^{bcddef}	375.50 ± 9.95 ^{ab}
P - value	0.000	0.96	0.000	0.003

Table 4. Anti-nutritional factors, total phenol, and total flavonoid content in *injera* samples. *CON1–100% sorghum flour *injera*, CON2–100% *teff* flour *injera*, BR – blending ratio (from BR1 up to BR10), Each value means ± standard deviation of triplicates. The values followed by different superscript letters within a column indicate a significant difference ($p < 0.05$).

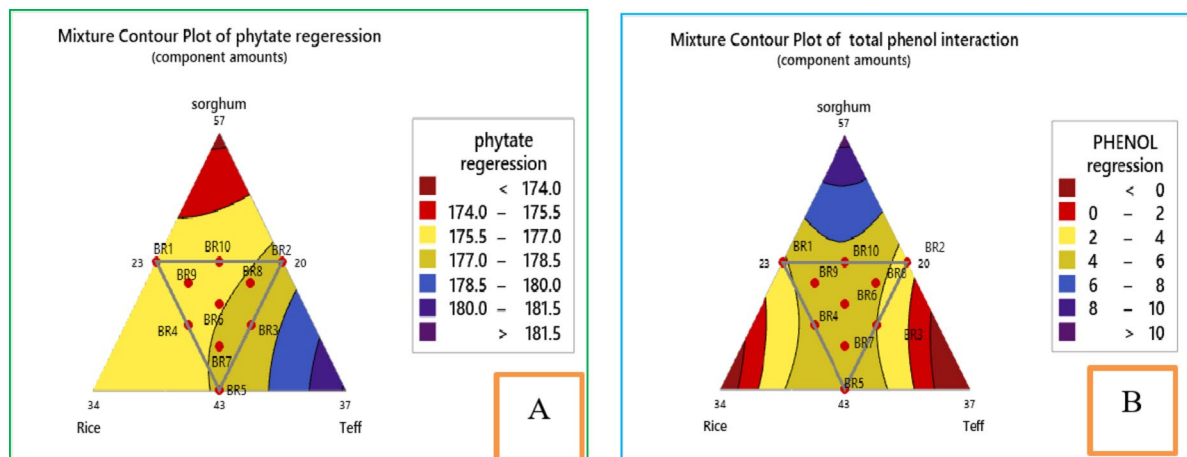


Fig. 4. Contour plot of phytate (A) and total phenol component (B) response.

Regression	Phytate content	Total phenol content	Condensed tannin content	Total flavanoid content
Linear	0.383	0.048	0.871	0.841
Quadratic	0.713	0.026	0.896	0.349
Sorghum*Rice	0.881	0.042	0.887	0.148
Sorghum*Teff	0.686	0.765	0.874	0.354
Rice*Teff	0.997	0.952	0.486	0.715
R ²	0.6366	0.8886	0.6893	0.7834

Table 5. ANOVA compositions p-values of anti-nutritional factor analysis.

emphasized or reported that white sorghum flour observed low in total phenol content of *injera* has been agreed with reported by^{63,64}, and another scientific justification reported studies blending, fermentation, and baking process parameters also reduce the anti-nutritional compounds and increased mineral bioavailability to the optimized nutritional quality of blended *injera*^{22,65}. The result shows a significant difference ($p < 0.05$) in the linear and quadratic model mixture interaction of sorghum with rice and sorghum with *teff* (Table 5).

$$TP = 1.53981 * \text{sorghum} - 3.26747 * \text{rice} - 3.33204 * \text{teff} - 0.01239 * \text{sorghum} * \text{rice} - 0.00256 * \text{sorghum} * \text{teff} - 0.17912 * \text{rice} * \text{teff} \quad (22)$$

The phytate content in *injera* made from blends aligns with the findings reported by^{11,43,51}. The phytate content in *injera* made from *teff* flour specifically also corresponds with the results reported by¹¹. The study showed no significant difference ($p < 0.05$) in the interaction between linear and quadratic models regarding the mixture of components, likely because all components in the mixture are rich sources of phytate. This anti-nutritional content chelates and binds minerals, reducing their bioavailability for nutritional absorption^{1,61}, but also provides health benefits, such as anti-inflammatory and anti-cancer properties^{30,65}. According to the model (Eq. 22) and the mixture contour plot (Fig. 5), the phytate content increases as the proportion of *teff* flour, followed by rice and sorghum, increases.

$$\text{CT model} = -0.093574 * \text{sorghum} + 0.045484 * \text{rice} - 0.143202 * \text{teff} + 0.001069 * \text{sorghum} * \text{rice} + 0.00480 * \text{sorghum} * \text{teff} - 0.000385 * \text{rice} * \text{teff} \quad (23)$$

The condensed tannin (CT) content in *injera* made from different composite flour ratios ranged from 0.00 to 0.17 mg/100 g. The control *injer*as made from sorghum and *teff* flour showed CT contents of 0.04 mg/100 g and 0.00 mg/100 g, respectively. A significant mean difference ($p > 0.05$) was observed between each blending ratio and the controls in terms of condensed tannin content, as shown in Table 4. Among the ten blended *injera* products, the highest condensed tannin content (0.17 mg/100 g) was found in BR8, which consisted of 48.83% sorghum, 22.34% rice, and 28.83% *teff*. This variation in CT content may be attributed to differences in fermentation composition⁶⁶. These findings align with previous studies^{11,63,67,68}, which have shown that white sorghum and *teff* flour have very low or no tannin content, respectively. The following contour plot (Fig. 5) and model (Eq. 23) also demonstrate that as the amount of brown rice increases, the concentration of condensed tannin increases, whereas an increase in the composition of white sorghum and *teff* flour leads to a decrease in condensed tannin concentration.

The total flavonoid (TF) content in the blended *injera* ranged from 298.5 to 388.2 mg/100 g. For comparison, the TF content in *injera* made from sorghum and *teff* flours alone was 289.9 mg/100 g and 304.9 mg/100 g, respectively. The highest TF content (388.2 mg/100 g) was observed in the BR6 blend (47.67% sorghum, 24.67%

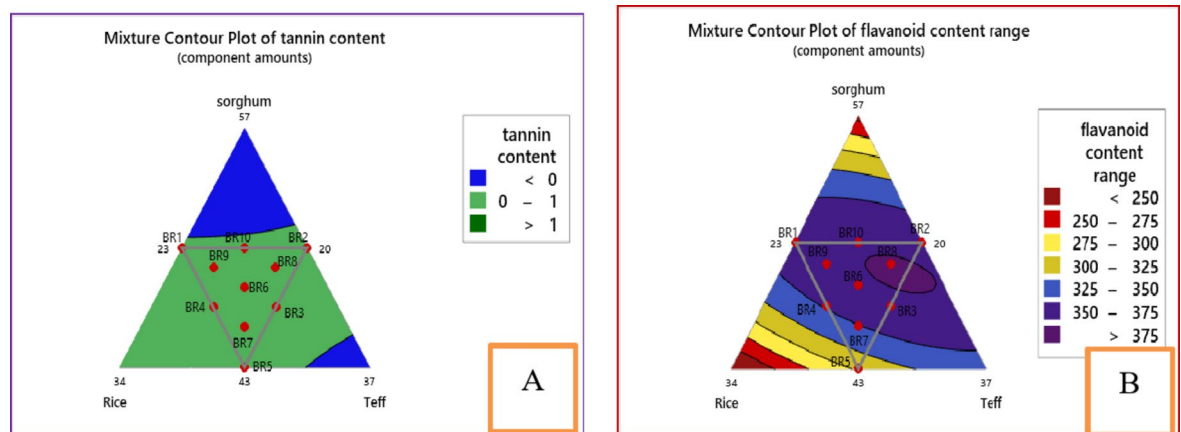


Fig. 5. Contour plot of condensed tannin (A) and flavanoid content (B) response.

rice, 27.66% teff), while the lowest TF content (289.9 mg/100 g) was found in the BR5 blend (43% sorghum, 27% rice, 30% teff). These findings suggest that the flavonoid content originates from the outer layer and, aligns with previous research on the flavonoid content of grains like sorghum, rice, teff, and flaxseed^{63,64,69–71}. Flavonoids are present in small amounts in sorghum, teff, rice, and flaxseed flours. Even though, the result indicates that in the mixture contour plot when sorghum and rice flour composition were increased the total flavonoid content also decreased in the blended *injera* foods but relatively increased by teff flour composition increased. According to researchers²², and⁶⁵ report fermentation and baking process reduced these anti-nutritional compounds and increased mineral bioavailability to the optimized nutritional quality of blended *injera*. There was a non-significant difference ($p < 0.05$) in the linear and quadratic model interaction on the mixture of components (Table 5).

$$TF = -48.3797 * \text{sorghum} - 90.6237 * \text{rice} - 30.8675 * \text{teff} + 2.7514 * \text{sorghum} * \text{rice} + 1.6083 * \text{sorghum} * \text{teff} + 0.6016 * \text{rice} * \text{teff} \quad (24)$$

Mineral content analysis of composite flour *injera*

In this study, the contents of iron, zinc, calcium, magnesium, manganese, and potassium in the blended *injera* were analyzed. The iron content in the formulated blended *injera* varied between 12.13 and 16.17 mg per 100 g. In comparison, *injera* made from 100% sorghum flour had 11.1 mg of iron per 100 g, while *injera* made from 100% teff flour contained 24.50 mg per 100 g. The iron content of the 100% sorghum flour *injera* (used as a control) and the blended *injera* did not show a significant difference ($p < 0.05$). However, the component interaction in the blended *injera* improved the iron content from 11.1 mg/100 g to a range of 12.13–16.17 mg/100 g. The highest iron content (16.17 mg/100 g) was found in the blend BR5 (43% sorghum, 27% rice, and 30% teff), while the lowest iron content (12.13 mg/100 g) was found in the blend BR1 (50% sorghum, 27% rice, and 23% teff).

The iron mineral content in teff flour significantly influences the iron content of *injera*. Pure teff flour *injera* serves as a control with an iron content of 24.50 mg/100 g, indicating that teff flour is richer in iron compared to *injera* made from other cereal blends^{36,50}. Specifically, *injera* made from 100% teff flour has a significantly higher iron content (25.13 mg/100 g to 34.50 mg/100 g, $p < 0.05$) than *injera* made from other cereal flour blends, due to the naturally high iron content of teff grain^{50,72}. Moreover, the iron content of blended *injera* is improved compared to *injera* made from 100% sorghum flour, which contains only 11.10 mg/100 g of iron.

In this study, the predicted (Equation model 25) and the mixture contour plot Fig. 6 shows that the iron content of composite flour blending *injera* optimization is influenced by sorghum flour > teff flour > rice flour. According to¹⁸, and⁷² reports the total iron contents of composite flour (teff, rice, maize, fenugreek) *injera* vary from (12.25–34.22 mg/100 g). The iron content in formulated *injera* showed high significance differently ($p < 0.05$) both in linear and quadratic models' interaction of sorghum*teff flour and rice*teff flour but the non-significant difference ($p < 0.05$) in sorghum*rice flour interaction (Table 6). Because it might be the iron content of sorghum and rice flour is non significantly different and not rich availability of iron content (Table 7).

$$Fe = 4.37493 * \text{sorghum} - 3.47793 * \text{rice} + 3.01581 * \text{teff} - 0.05511 * \text{sorghum} * \text{rice} - 0.17160 * \text{sorghum} * \text{teff} + 0.14472 * \text{rice} * \text{teff} \quad (25)$$

The Calcium content of formulated *injera* was found in a range of 1.07–4.06 mg/100 g, 100% sorghum flour *injera* (1.07 mg/100 g), and 100% teff flour (3.14 mg/100 g), respectively. The highest calcium content (4.06 mg/100 g) was found in BR1 (50% sorghum, 27% rice, and 23% teff) whereas, the lowest calcium content (1.07 mg/100 g) was found in BR7 (45.34% sorghum, 25.83% rice, and 28.83% teff) within constant 3% supplement of flaxseed flour for each blended *injera* products. The calcium content of formulated *injera* varied and lower from previous studies, it might be due to variations in composition amount, composition type, and any factors that contributed to calcium content.

On the other hand, the lower amount of calcium might be due to divalent cations, such as Ca, binding with anti-nutritional factors, which could explain the reduced availability of these minerals. Various researchers have reported that the calcium content in composite flour *injera* ranges from 17.97 to 187.25 mg/100g^{11,32,72,73}. In this

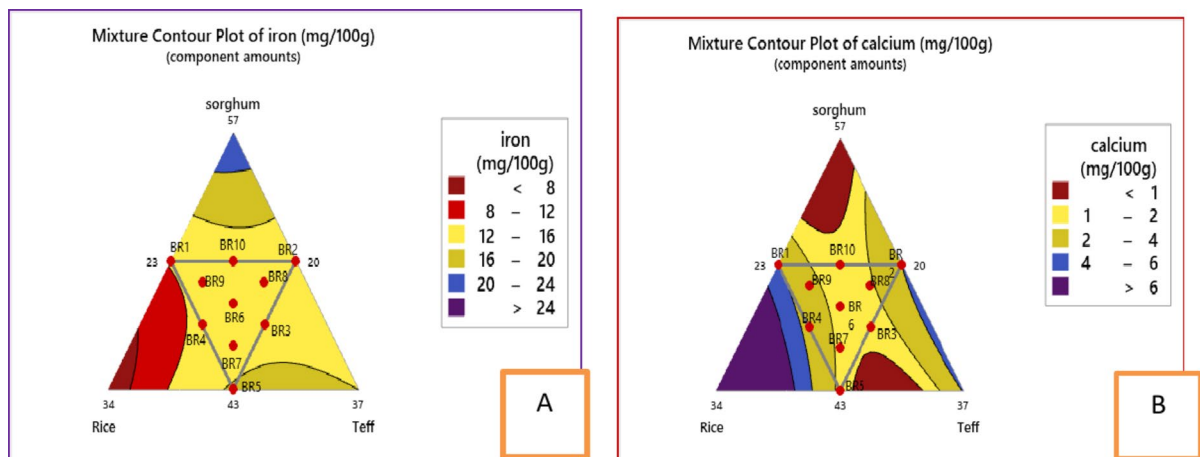


Fig. 6. Contour plot of iron (A) and calcium content component (B) response.

Regression	Iron	Zinc	Calcium	Magnesium	Manganese	Potassium
Linear	0.006	0.042	0.220	0.194	0.023	0.033
Quadratic	0.004	0.036	0.102	0.025	0.016	0.039
Sorghum*Rice	0.097	0.034	0.694	0.017	0.397	0.147
Sorghum*Teff	0.003	0.858	0.027	0.049	0.006	0.152
Rice*Teff	0.005	0.020	0.448	0.057	0.044	0.015
R ²	0.978	0.8683	0.8238	0.9184	0.9157	0.8940

Table 6. ANOVA for mixture compositions p-values of mineral analysis.

Composition	Fe content (mg/100 g)	Zn content (mg/100 g)	Ca content (mg/100 g)	Mg content (mg/100 g)	Mn content (mg/100 g)	K content (mg/100 g)
CON1	11.10 ± 3.23 ^b	2.16 ± 1.45 ^b	1.07 ± 0.31 ^a	5.31 ± 0.10 ^a	2.50 ± 0.01 ^d	38.67 ± 0.34 ^b
CON2	24.50 ± 2.36 ^a	3.49 ± 0.02 ^a	3.14 ± 1.25 ^a	6.81 ± 0.12 ^a	6.76 ± 0.48 ^a	40.33 ± 0.39 ^{ab}
BR1	12.13 ± 3.24 ^b	3.94 ± 0.01 ^a	4.06 ± 1.49 ^a	6.73 ± 0.80 ^a	6.05 ± 0.69 ^{abc}	38.88 ± 0.40 ^b
BR2	14.12 ± 2.64 ^b	3.92 ± 0.01 ^a	4.00 ± 2.59 ^a	6.43 ± 0.80 ^a	7.24 ± 1.69 ^a	40.90 ± 0.37 ^{ab}
BR3	14.49 ± 3.36 ^b	3.33 ± 0.00 ^{ab}	1.68 ± 0.28 ^a	5.96 ± 0.14 ^a	4.37 ± 0.12 ^{bcd}	41.22 ± 0.36 ^{ab}
BR4	12.36 ± 0.63 ^b	2.16 ± 0.06 ^b	3.76 ± 1.26 ^a	6.48 ± 0.11 ^a	2.62 ± 0.22 ^d	40.03 ± 1.84 ^{ab}
BR5	16.57 ± 3.32 ^{ab}	3.27 ± 0.01 ^{ab}	1.79 ± 0.00 ^a	5.55 ± 0.12 ^a	6.27 ± 0.73 ^{ab}	41.19 ± 0.37 ^{ab}
BR6	14.22 ± 1.62 ^b	3.37 ± 0.01 ^{ab}	1.94 ± 0.35 ^a	5.68 ± 0.15 ^a	3.75 ± 0.67 ^d	42.73 ± 0.27 ^a
BR7	14.54 ± 2.71 ^b	2.76 ± 0.01 ^{ab}	1.07 ± 0.31 ^a	6.27 ± 1.13 ^a	4.08 ± 0.88 ^{cd}	41.17 ± 0.26 ^{ab}
BR8	14.15 ± 4.57 ^b	2.98 ± 0.00 ^{ab}	1.61 ± 0.31 ^a	5.47 ± 1.50 ^a	3.50 ± 0.71 ^d	41.05 ± 0.28 ^{ab}
BR9	13.20 ± 1.36 ^b	3.16 ± 0.01 ^{ab}	2.16 ± 0.32 ^a	5.58 ± 0.12 ^a	4.01 ± 0.02 ^{cd}	41.47 ± 0.28 ^{ab}
BR10	14.85 ± 2.11 ^b	3.49 ± 0.55 ^{ab}	1.61 ± 0.31 ^a	6.00 ± 1.17 ^a	2.92 ± 0.07 ^d	43.15 ± 3.11 ^a
P-value	0.001	0.001	0.066	0.187	0.000	0.001

Table 7. Mineral content of sorghum, rice, and *teff* composite flour *injera*. *CON1- 100% sorghum flour *injera*, CON2-100% *teff* flour *injera*, Fe - iron, Zn-zinc, Ca – calcium, Mg – magnesium, Mn – manganese, K – potassium. Each value means ± standard deviation of triplicates. The values followed by different superscript letters within a column indicate a significant difference ($p < 0.05$).

study, the mean value of blended *injera* products did not show a significant difference, but the calcium content was optimized. As the amount of rice increased, followed by *teff* and sorghum flour, the calcium content also increased. This suggests that the contribution of flaxseed flour to calcium content remained constant across all blended experimental trials (Fig. 6). The calcium content was found to be significantly different ($p < 0.05$) in the linear and quadratic model interactions of sorghum-rice flour and rice-*teff* flour. However, there was no significant difference ($p > 0.05$) in the interaction between sorghum and *teff* (Table 6).

$$Ca = 0.9540 * \text{sorghum} + 10.7782 * \text{rice} + 1.9126 * \text{teff} - 0.1753 * \text{sorghum} * \text{rice} - 0.0106 * \text{sorghum} * \text{teff} - 0.2090 * \text{rice} * \text{teff} \quad (26)$$

The zinc contents of the formulated *injera* were found in a range of 2.16–3.94 mg/100 g, 100% sorghum flour *injera* (2.16 mg/100 g), and 100% *teff* flour *injera* (3.49 mg/100 g) respectively. The *injera* made from BR1 (containing 50% sorghum, 27% rice, and 23% *teff*) had the highest zinc content at 3.94 mg per 100 g, while BR4 (comprising 46.5% sorghum, 27% rice, and 26.5% *teff*) contained the lowest at 2.16 mg per 100 g. The result showed that Zn concentration was also optimized or increased by blending different grains (Table 6). According to^{13,59,72} studied composite flour *injera* made from different cereals at different composition levels zinc content varied from (1.44 to 4.35 mg/100 g), and the present study result agreement with the researchers.

The mixture contour plot (Fig. 7) and prediction model Eq. 27 shows that sorghum and *teff* flour composition increases its Zn content also but as rice composition increases Zn content value decreases.

$$Zn = 1.58405 * \text{sorghum} - 0.50890 * \text{rice} + 3.87093 * \text{teff} - 0.01272 * \text{sorghum} * \text{rice} - 0.10232 * \text{sorghum} * \text{teff} - 0.02531 * \text{rice} * \text{teff} \quad (27)$$

The magnesium content of formulated *injera* ranged from 5.47 to 6.73 mg/100 g. Specifically, 100% sorghum flour *injera* contained 5.31 mg/100 g, while 100% *teff* flour *injera*, used as a control, had 6.81 mg/100 g, as shown in Table 7. The highest magnesium content (6.73 mg/100 g) was found in the BR1 formulation (50% sorghum, 27% rice, and 23% *teff*), whereas the lowest (5.47 mg/100 g) was in the BR8 formulation (48.83% sorghum, 22.34% rice, and 28.83% *teff*). According to⁷³, composite flour *injera* made from *teff* and other functional food ingredients had magnesium content ranging from 6.4 to 14.05 mg/100 g, which aligns with the results of the present study. The magnesium content showed no significant difference ($p < 0.05$) in the interaction of rice and *teff* flour in both linear and quadratic models, but there was a significant difference ($p < 0.05$) in the interaction of sorghum and rice flour, and sorghum and *teff* flour in the quadratic model (Table 7). The interaction composition effect of Mg is predicted as:

$$Mg = 1.78941 * \text{sorghum} + 4.07668 * \text{rice} + 2.42312 * \text{teff} - 0.09649 * \text{sorghum} * \text{rice} - 0.06802 * \text{sorghum} * \text{teff} - 0.06461 * \text{rice} * \text{teff} \quad (28)$$

The equation (Eq. 28) implies that as the mixture counter plot shown the composition of rice and sorghum flour in the blending increases the magnesium mineral concentration in the product. The supplemented flaxseed flour also increased the Mg concentration in the formulated *injera*. The manganese content of formulated *injera* was found to range from 2.92 to 7.24 mg/100 g. For comparison, sorghum flour *injera* (used as a control) contained 2.50 mg/100 g of manganese, while *teff* flour *injera* (another control) contained 6.76 mg/100 g (Table 7). There has been limited research on the manganese mineral content in composite flour *injera* products. The mixture contour plot (Fig. 8) and the prediction model (Eq. 29) indicated that increasing the proportions of sorghum and *teff* in the composition led to a higher manganese content. Although rice flour is also an important source of manganese, its role in enhancing manganese availability was not as significant^{53,74}.

$$Mn = 6.97845 * \text{sorghum} - 6.20752 * \text{rice} + 7.29580 * \text{teff} - 0.05716 * \text{sorghum} * \text{rice} - 0.32571 * \text{sorghum} * \text{teff} + 0.17489 * \text{rice} * \text{teff} \quad (29)$$

The potassium content of formulated *injera* ranged from 38.88 to 43.15 mg/100 g. Specifically, *injera* made from 100% sorghum flour had 38.67 mg/100 g, and *injera* made from 100% *teff* flour had 40.33 mg/100 g (Table 7). The highest potassium content (43.15 mg/100 g) was found in the BR10 blend (50% sorghum, 23.5% rice, and 26.5% *teff*), while the lowest (38.88 mg/100 g) was found in the BR1 blend (50% sorghum, 27% rice, and 23% *teff*). According to⁷⁵, rice is a common source of potassium. However, the mixture contour plot showed that increasing rice composition led to a linear decrease in potassium content in the formulated *injera*. There are limited studies on the potassium content in composite flour *injera* made from cereal grains. The mixture contour plot and prediction model indicated that the potassium content is primarily affected by the sorghum composition in the blend (Fig. 8). A significant difference ($p < 0.05$) was observed in the linear and quadratic model interactions of rice and *teff* flour, whereas the quadratic model interactions of sorghum with rice and sorghum with *teff* flour showed no significant difference (Table 6).

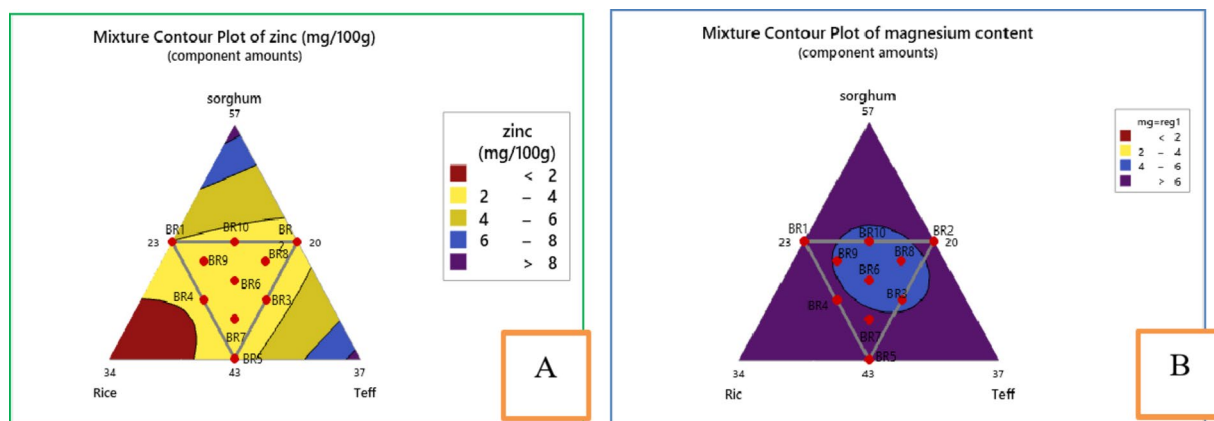


Fig. 7. Contour plot of zinc (A) and magnesium content (B) interaction response.

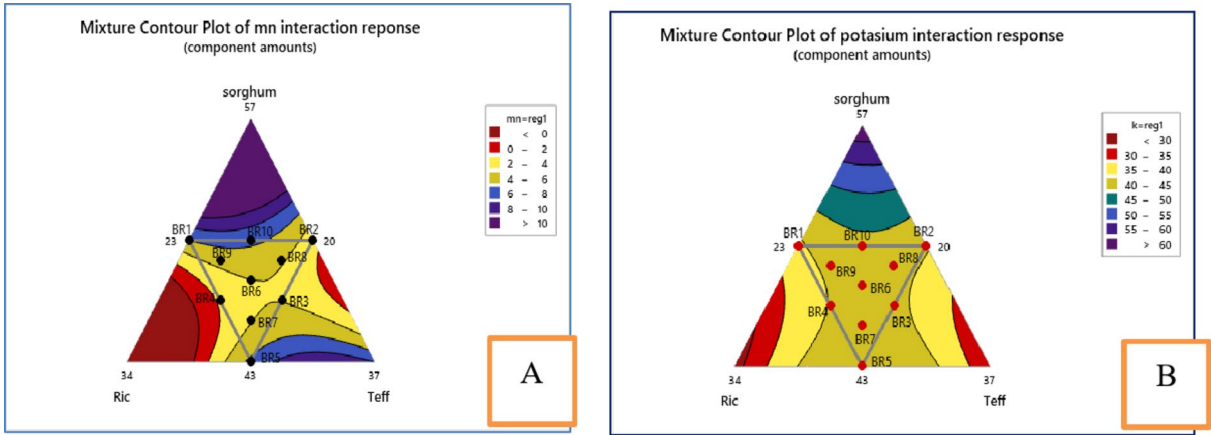


Fig. 8. Contour plot of manganese (A) and potassium component (B) response.

Composition	Yeast-mold load				Total bacteria load			
	Day 0 log (cfug ⁻¹)	Day 2 log (cfug ⁻¹)	Day 4 log (cfug ⁻¹)	Day 6 log (cfug ⁻¹)	Day 0 log (cfug ⁻¹)	Day 2log (cfug ⁻¹)	Day 4log (cfug ⁻¹)	Day 6log (cfug ⁻¹)
CON1	ND	2.61 ± 0.10 ^a	3.59 ± 0.03 ^a	2.48 ± 0.01 ^b	ND	2.48 ± 0.01 ^a	3.59 ± 0.03 ^a	2.49 ± 0.02 ^b
CON2	ND	2.63 ± 0.01 ^a	3.68 ± 0.03 ^a	4.57 ± 0.05 ^a	ND	2.59 ± 0.03 ^a	3.64 ± 0.01 ^a	4.59 ± 0.09 ^a
BR1	ND	2.50 ± 0.02 ^a	3.62 ± 0.04 ^a	4.60 ± 0.04 ^a	1.24 ± 0.75 ^a	2.43 ± 0.08 ^a	3.55 ± 0.06 ^a	4.61 ± 0.02 ^a
BR2	ND	2.49 ± 0.04 ^a	3.59 ± 0.02 ^a	4.71 ± 0.17 ^a	ND	2.47 ± 0.06 ^a	3.56 ± 0.02 ^a	4.63 ± 0.07 ^a
BR3	ND	2.50 ± 0.05 ^a	3.61 ± 0.02 ^a	4.69 ± 0.03 ^a	ND	2.49 ± 0.02 ^a	3.60 ± 0.02 ^a	4.61 ± 0.02 ^a
BR4	ND	2.52 ± 0.12 ^a	3.63 ± 0.05 ^a	4.62 ± 0.05 ^a	1.31 ± 0.85 ^a	2.48 ± 0.07 ^a	3.53 ± 0.01 ^a	4.61 ± 0.02 ^a
BR5	ND	2.52 ± 0.04 ^a	3.59 ± 0.02 ^a	4.65 ± 0.08 ^a	1.25 ± 0.76 ^a	2.51 ± 0.05 ^a	3.59 ± 0.02 ^a	4.56 ± 0.03 ^a
BR6	ND	2.55 ± 0.07 ^a	3.61 ± 0.06 ^a	4.59 ± 0.03 ^a	1.26 ± 0.78 ^a	2.52 ± 0.2 ^a	3.58 ± 0.03 ^a	4.61 ± 0.03 ^a
BR7	ND	2.51 ± 0.10 ^a	3.55 ± 0.05 ^a	4.60 ± 0.10 ^a	ND	2.49 ± 0.04 ^a	3.55 ± 0.12 ^a	4.62 ± 0.01 ^a
BR8	ND	2.51 ± 0.03 ^a	3.60 ± 0.02 ^a	4.61 ± 0.00 ^a	ND	2.49 ± 0.09 ^a	3.53 ± 0.05 ^a	4.63 ± 0.07 ^a
BR9	ND	2.52 ± 0.04 ^a	3.61 ± 0.00 ^a	4.59 ± 0.02 ^a	ND	2.50 ± 0.03 ^a	3.56 ± 0.08 ^a	4.61 ± 0.02 ^a
BR10	ND	2.48 ± 0.05 ^a	3.61 ± 0.03 ^a	4.62 ± 0.22 ^a	ND	2.48 ± 0.04 ^a	3.59 ± 0.03 ^a	4.61 ± 0.01 ^a
P-value (BR factor)	–	0.044	0.044	0.044	0.009	0.009	0.009	0.009
P-value (SD factor)	–	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Table 8. The microbial load of composite flour *injera* at different storage durations. *BR – blending ratio, SD – storage duration, CON1-100% sorghum flour, CON2-100% *teff* flour, ND – not detected Each value means ± standard deviation of duplicates, the values followed by different superscript letters within a column indicate a significant difference ($p < 0.05$).

$$K = 6.52677 \times \text{sorghum} - 3.24099 \times \text{rice} - 2.18414 \times \text{teff} - 0.13914 \times \text{sorghum} \times \text{rice} - 0.13677 \times \text{sorghum} \times \text{teff} + 0.31492 \times \text{rice} \times \text{teff} \quad (30)$$

Microbial load analysis of *injera* at different storage periods

The effects of blending ratios and flaxseed supplementation on the microbial quality of *injera* samples over storage durations of 0, 2, 4, and 6 days after baking are presented in Table 8. On day 0, no yeast or mold was detected. By the second day of storage, yeast and mold colonies were present in the 100% sorghum and *teff* flour *injera* used as controls, while the ten experimental trials of blended *injera* from sorghum, rice, and *teff* composite flours were supplemented with 3% flaxseed flour. The highest yeast and mold counts (2.55 cfu/g) and the lowest (2.48 cfu/g) were observed in the day 2 storage duration for *injera* with whole flaxseed flour supplementation. In the 100% sorghum and *teff* flour *injera*, yeast and mold counts were 2.61 cfu/g and 2.63 cfu/g, respectively. On the fourth day of storage, the highest yeast and mold colony count was 3.63 cfu/g, and the lowest was 3.55 cfu/g. For 100% sorghum and *teff* flour *injera*, the counts were 3.59 cfu/g and 3.68 cfu/g, respectively. By the sixth day of storage, the highest yeast and mold colony count was 4.71 cfu/g, and the lowest was 4.59 cfu/g for all blended *injera*. The 100% sorghum and *teff* flour *injera* had yeast and mold colony counts of 4.48 cfu/g and 4.57 cfu/g, respectively. There was no significant ($p < 0.05$) difference in yeast and mold colony counts across different storage periods. However, by the sixth day, the yeast and mold colony count in sorghum flour *injera* had decreased compared to the 100% *teff* flour *injera* and blended *injera* products (composite flour with flaxseed flour supplement). This reduction is attributed to the loss of moisture content, which minimizes favorable conditions for microbial growth.

According to the study by⁷⁶, flaxseed demonstrated significant antifungal properties, which contributed to the observed reduction in yeast mold and total bacterial microbial load. This might be due to the relatively low concentration of *teff* in the blends. In contrast, the microbial load in *injera* made from sorghum and *teff* flour was higher than in the flaxseed blends. The yeast-mold colony count was significantly affected by both the blending ratio (BR) and the storage period (SP) ($p < 0.05$). Reference⁷² reported microbial yeast-mold colony counts ranging from 1.93 to 4.02 CFU/g and total bacterial colony counts from 2.63 to 4.02 CFU/g in *teff* and fenugreek composite flour *injera* over different storage times. The current study's findings align with⁷²'s results. Incorporating flaxseed flour into *injera* led to a decrease in microbial counts, as both yeast mold and total bacterial counts were highest in the control samples of *injera*, corroborating the findings of²². Figure 9 displays the blending ratios of *injera* products from 12 experimental trials alongside air-dried *injera*, prepared for further analysis.

Total bacterial growth was observed in some blending *injera* sample products during day 0 enumeration, likely due to material contamination, as the baked *injera* had undergone thermal processing. On the 2nd day of storage, the highest colony count was 2.52 cfu/g, and the lowest was 2.43 cfu/g. Similarly, the microbial quality of *injera* made from 100% sorghum flour (2.48 cfu/g) and 100% *teff* flour (2.59 cfu/g) showed slightly higher total bacterial loads. By the 4th day of storage, the highest total bacterial growth observed was 3.60 cfu/g, and the lowest was 3.53 cfu/g. The control samples of sorghum flour and *teff* flour *injera* had bacterial colony counts of 3.59 cfu/g and 3.64 cfu/g, respectively. On the 6th day, the highest and lowest total bacterial loads observed were 4.63 cfu/g and 4.56 cfu/g, respectively. The control bacterial colony counts for sorghum and *teff* flour *injera* were 2.49 cfu/g and 4.59 cfu/g, respectively. This increase in bacterial load is attributed to the staling effect in sorghum flour *injera*, caused by moisture transfer from crumb to crust and intrinsic firming of the cell wall material, which is associated with starch recrystallization during storage⁷⁷. The growth of total bacterial colonies was evident by the 6th day of storage.

Optimization of overall acceptable quality *injera*

Acceptability of the overall optimal formulation value to develop *injera* having high qualities about its proximate, mineral, and combination of their response values. The nutritional quality of *injera* covered all the main important parameters, those of its proximate and mineral concentration due to its better quality for consumers. BR2 formulation was labeled to be optimized which was in the ratio of 50% (sorghum), 20% (rice), and 30% (*teff*), the optimized blended ratio based on macro-nutritional (proximate), micro-nutritional (mineral) with the desirability of 0.9474. Table 9 displays the results of the overall optimized response parameters.

Optimization of the proximate composition of *injera* for protein, fiber, carbohydrate, and energy increases progressively as the composition of sorghum in the component increases with constant flaxseed flour supplemented for each blending. The white region in (Fig. 10) shows that an optimum proximate content, optimum mineral content, of the formula of sorghum, *teff*, and rice supplemented with flaxseed flour results in desirable attributes.

The optimum responses of fiber, protein, carbohydrate, energy, iron, zinc, and calcium contents were 3.02%, 9.30%, 83.16%, 393.24 kcal/100 g, 13.99 mg/100 g, 3.78 mg/100 g, and 4.06 mg/100 g, respectively. The total mineral content of *injera* progressively increases due to the blending ratio as compared to controls. Optimization suggested that *injera* made with 50% sorghum, 20% rice, and 30% *teff* achieved the best solution for this combination of variables with a desirability of 0.9474.



Fig. 9. The 12 experimental trial blending ratio *injera* products (The resulting product represented (1) Control 1, (2) Control 2, (3) BR1, (4) BR2, (5) BR3, (6) BR4, (7) BR5, (8) BR6, (9) BR7, 10. BR8, 11. BR9, and 12. BR10.

Optimized parameter	Goal	Lower	Target	Upper	Desirability	Composite desirability
Crude fiber (%)	Maximum	2.4	3	3	1	0.947445
Crude protein (%)	Maximum	9	9.2	9.2	1	
Carbohydrate (%)	Maximum	81.39	83	83	1	
Energy (kcal/100 g)	Maximum	388.4	390	390	1	
Overall acceptability	Maximum	2.33	4	4	1	
Iron (mg/100 g)	Maximum	12.13	15	15	1	
Zinc (mg/100 g)	Maximum	2.16	3	3	1	
Calcium (mg/100 g)	Maximum	1.07	3	3	1	

Table 9. Overall optimized response parameters result.

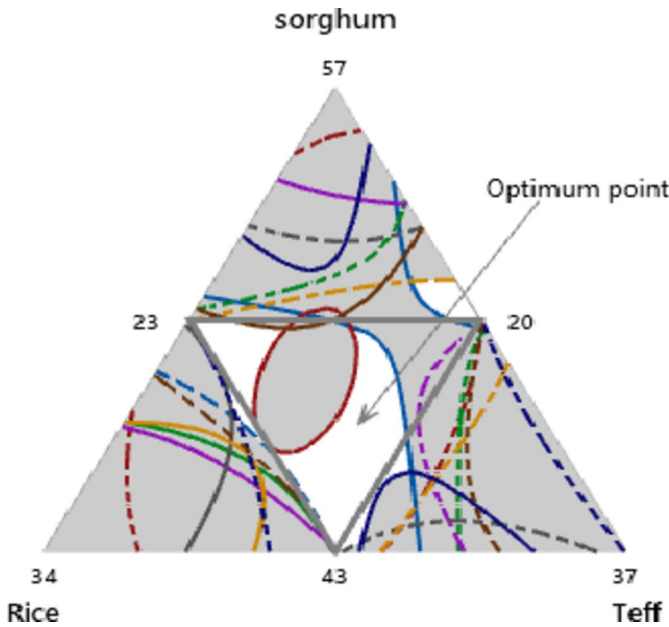


Fig. 10. Overlaid mixture contour plot of overall optimization.

Conclusion

A nutritionally enhanced injera was produced by blending sorghum, rice, and teff flours with a small amount of flaxseed flour. The study analyzed the proximate, functional, and mineral composition of the flours, anti-nutritional factors, and microbial load to improve injera’s nutritional quality compared to a control made from 100% sorghum flour. The optimal blend, consisting of 50% sorghum, 20% rice, 30% teff, and 3% flaxseed flour, resulted in injera with 3.02% fiber, 9.30% protein, 83.16% carbohydrates, 393.24 kcal/100 g energy, and notable levels of iron (13.99 mg/100 g), calcium (4.06 mg/100 g), and zinc (3.78 mg/100 g). This blend was identified as the most nutritionally beneficial.

The highest crude fat content (3.20%) was found in injera made with a 50% sorghum, 20% rice, and 30% teff blend (BR2). The highest crude fiber content (4.50%) was observed in injera made from BR4 (46.5% sorghum, 27% rice, 26.5% teff) and BR5 (43% sorghum, 27% rice, 30% teff). The highest crude protein (9.31%) and carbohydrate (83.67%) contents were also in BR2. Regarding anti-nutritional factors, the highest phytate content (177.93 mg/100 g) was found in BR7 (45.34% sorghum, 25.83% rice, 28.83% teff), total phenols (5.39 mg/100 g) in BR2, total flavonoids (388.2 mg/100 g) in BR6 (47.67% sorghum, 24.67% rice, 27.66% teff), and condensed tannins (0.17 mg/100 g) in BR8 (48.83% sorghum, 22.34% rice, 28.83% teff). The mineral content improved with the blend of sorghum, rice, and teff, suggesting the potential of underutilized grains and seeds to enhance the nutritional value of injera. We recommend applying this approach to other traditional and modern food products, such as bread, porridges, and snack bars, incorporating diverse grain and seed flours. This could help address nutritional deficiencies and improve the dietary profiles of populations that rely on staple foods.

Data availability

The data used to support the findings of this study are included in the article.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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